



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Computer Science and Engineering Department

Name of the teacher: Rashmi Pal		Hours/Week: 3
Subject Name: 303105251 - Operating System		Hours Mon, Wed, Sat
Sr. No.	Name of Topic	Planned Date
1.	Concept of Operating Systems, Generations of Operating systems	19/01/26
2.	OS Services, System calls, Structure of an OS-Layered, Monolithic	19/01/26
3.	Concept of Virtual Machine	19/01/26
4.	Definition, Process Relationship, Different states of a Process, Process State transitions	19/01/26
5.	Process Control Block (PCB), Context switching	22/01/26
6.	Thread: Definition, Various states, Benefits of threads, Types of threads,	24/01/26
7.	Concept of multithreads. Process Scheduling: Foundation	02/02/26
8.	Types of Schedulers, Scheduling criteria: CPU utilization, Throughput.	02/02/26
9.	Scheduling algorithms: Pre-emptive and Non pre-emptive,	05/02/26
10.	FCFS	09/02/26
11.	SJF	09/02/26
12.	RR	12/02/26
13.	Inter-Process Communication: Critical Section, Race Conditions	16/02/26
14.	Hardware Solution, Strict Alternation, Peterson's Solution	19/02/26
15.	The Producer\ Consumer Problem, Semaphores, Event Counters, Monitors	23/02/26
16.	Classical IPC Problems: Reader's & Writer Problem,	23/02/26
17.	Dinning Philosopher Problem etc.	26/02/26
18.	Deadlocks: Definition, Necessary and sufficient conditions for Deadlock,	02/03/26
19.	Deadlock Prevention	05/03/26

20.	Deadlock Avoidance: Banker's algorithm	09/03/26
21.	Deadlock detection	09/03/26
22.	Deadlock Recovery	12/03/26
23.	Memory Management: Basic concept, Logical and Physical address map	16/03/26
24.	Memory allocation: Contiguous Memory allocation	16/03/26
25.	Fixed and variable partition	19/03/26
26.	Internal and External fragmentation and Compaction	30/03/26
27.	Paging: Principle of operation	30/03/26
28.	Page allocation	02/04/26
29.	Hardware support for paging, Protection and sharing,	06/04/26
30.	Virtual Memory: Basics of Virtual Memory	06/04/26
31.	Hardware and control structures, Locality of reference, Page fault	09/04/26
32.	Working Set , Dirty page/Dirty bit, Demand paging	13/04/26
33.	Page Replacement algorithms: Optimal	13/04/26
34.	First in First Out(FIFO), Second Chance (SC)	16/04/26
35.	Not recently used (NRU) and Least Recently used (LRU).	20/04/26
36.	I/O Hardware: I/O devices, Device controllers,	20/04/26
37.	Direct memory access, Principles of I/O Software: Goals of Interrupt handlers	23/04/26
38.	Device drivers, Device independent I/O software	27/04/26
39.	File Management: Concept of File, Access methods, File types	27/04/26
40.	File operation, Directory structure, File System structure	30/04/26
41.	Allocation methods (contiguous, linked, indexed) , Free-space management	04/05/26
42.	Directory implementation (linear list, hash table), efficiency and performance	04/05/26
43.	Disk Management: Disk structure, Disk scheduling algorithms -FCFS, SSTF,SCAN, C SCAN	04/05/26
44.	Disk reliability, Disk formatting	07/05/26
45.	Boot-block, Bad blocks	07/05/26